ASSOCIATED PRIMES OF A TENSOR PRODUCT WITH A DIRECT SUM OF FINITE MODULES OVER A LOCAL RING

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- Nakayama's lemma. Algebra, label algebra-lemma-NAK, invoked at proof line 4407. Upgrades the surjectivity of the endomorphism f' obtained from the displayed identity to bijectivity, which is the contradiction step.
- An endomorphism of a finite module that becomes an isomorphism after tensoring. Algebra, label algebra-lemma-fun, invoked at proof line 4409. Turns the surjectivity of f' into the statement that f' is an isomorphism, whence the induced map on the tensor product is an isomorphism, contradicting the annihilation of the chosen element.

Source reference. Original lemma 00DV: sources/stacks-chow/tags.txt, source lines 512–512. Navigation pointer for \ref{algebra-lemma-NAK} used inside the counted span (line 4407); the argument that consumes it is retained in the span.

Source reference. Original lemma 05G8: sources/stacks-chow/tags.txt, source lines 7052–7052. Navigation pointer for \ref{algebra-lemma-fun} used inside the counted span (line 4409); the argument that consumes it is retained in the span.

Lemma 0.1. *Let R be a local ring with maximal ideal $\mathfrak{m}$. Let $R \to S$ be a ring map. Let N be an S-module. Assume N is isomorphic as an R-module to a direct sum of finite R-modules. Then for any R-module M and for any prime $\mathfrak{q} \subset S$ which is an associated prime of $N \otimes_R M$ we have $\mathfrak{q} + \mathfrak{m}S \neq S$.*

Proof. Write $N = \bigoplus_{i \in I} M_i$ with each M_i a finite R-module. Let M be an R-module and let $\mathfrak{q} \subset S$ be an associated prime of $N \otimes_R M$ such that $\mathfrak{q} + \mathfrak{m}S = S$. Let $z \in N \otimes_R M$ be an element with $\mathfrak{q} = \mathrm{Ann}_S(z)$. After modifying the direct sum decomposition a little bit we may assume that $z \in M_1 \otimes_R M$ for some element

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$1 \in I$. Write $1 = f + \sum x_j g_j$ for some $f \in \mathfrak{q}$, $x_j \in \mathfrak{m}$, and $g_j \in S$. For any $g \in S$ denote g' the R-linear map

$$M_1 \to N \xrightarrow{g} N \to M_1$$

where the first arrow is the inclusion map, the second arrow is multiplication by g and the third arrow is the projection map. Because each $x_j \in R$ we obtain the equality

$$f' + \sum x_j g'_j = \mathrm{id}_{M_1} \in \mathrm{End}_R(M_1)$$

By Nakayama's lemma (Algebra, Lemma 00DV) we see that f' is surjective, hence by Algebra, Lemma 05G8 we see that f' is an isomorphism. In particular the map

$$M_1 \otimes_R M \to N \otimes_R M \xrightarrow{f} N \otimes_R M \to M_1 \otimes_R M$$

is an isomorphism. This contradicts the assumption that $fz = 0$. $\square$